

Week 6: Multi-rate DSP — Homework and Weekly Quiz

Downsampling • Decimation • Upsampling • Interpolation • Rational Rate Changes • CNN Strides

Instructor packet with solutions

Textbook alignment: Oppenheim & Schaffer §4.6; ML connection to convolutional architectures in Raschka.

Homework

The problems intentionally progress from direct rate-change calculations to design reasoning and ML connections. Problems 1–4 are suitable for all students; Problems 5–8 add senior-level synthesis.

Problem 1. Downsampling a finite sequence (10 pts)

Let $x[n]$ for $n = 0, \dots, 11$ be $\{3, 1, -2, 4, 0, 5, 2, -1, 6, 3, 0, 2\}$. Downsample by $M = 3$.

(a) List $x_d[n]$. (b) If the original sample rate is 12 kHz, what is the output sample rate? (c) Which original indices are discarded?

Problem 2. Maximum alias-free decimation factor (10 pts)

A discrete-time signal is known to be zero outside $|\omega| \leq 0.18\pi$. Assume an ideal anti-aliasing filter that must preserve the entire occupied band.

(a) Find the largest integer decimation factor M that can be used without aliasing. (b) What is the ideal cutoff boundary for that M ?

Problem 3. Where does a sinusoid alias? (12 pts)

The input is $x[n] = \cos(0.76\pi n)$ and the signal is downsampled by $M = 2$ without filtering.

(a) Find the output digital frequency in the principal interval $[-\pi, \pi)$. (b) Write an equivalent cosine for the output. (c) Explain why no post-filter can recover the original 0.76π component from the output alone.

Problem 4. Designing a decimator (14 pts)

You must decimate a sensor signal by $M = 4$. The useful input band extends to 0.18π and frequencies above 0.25π are known to be interference.

(a) Draw the block diagram. (b) State the ideal aliasing boundary π/M . (c) Propose reasonable FIR passband and stopband edges. (d) Explain why a linear-phase FIR can be attractive and name one case where an IIR or minimum-phase filter might be preferable.

Problem 5. Strided convolution as a rate changer (14 pts)

Consider $x = [1, 2, 0, -1, 3, 1, 0]$ and a length-3 kernel $h = [1/4, 1/2, 1/4]$. Use the deep-learning cross-correlation convention (do not reverse h), no padding, and stride 2.

(a) Compute all stride-1 valid outputs. (b) Keep every second result to obtain the stride-2 output. (c) Explain why this is equivalent to filtering followed by downsampling. (d) Is h a plausible anti-aliasing kernel? Justify qualitatively.

Problem 6. Upsampling and interpolation gain (12 pts)

Let $x[n] = \{1, 2, -1, 0\}$ and upsample by $L = 3$.

(a) Write the zero-inserted sequence. (b) How many spectral images occur over one 2π DTFT interval? (c) What ideal lowpass cutoff should the interpolation filter use? (d) What DC gain should the filter have, and why?

Problem 7. Rational sample-rate conversion (14 pts)

Convert a stream from 30 kHz to 20 kHz.

(a) Reduce the rate ratio to L/M . (b) Draw the conceptual $\uparrow L \rightarrow H(z) \rightarrow \downarrow M$ chain. (c) Find the ideal combined-filter cutoff on the high-rate intermediate normalized-frequency axis. (d) State the required DC gain of the interpolation/combined filter. (e) Give one reason a polyphase implementation is preferred over explicitly inserting all zeros.

Problem 8. Transposed convolution and periodic artifacts (14 pts) — Challenge

A 1-D generator uses `ConvTranspose1d` with stride 2 and kernel length 5.

(a) Explain the zero-insertion + filtering interpretation. (b) Why can kernel length 5 with stride 2 produce nonuniform overlap across output positions? (c) Give two mitigation strategies. (d) Explain how an anti-imaging-inspired initialization can help without guaranteeing artifact-free outputs.

Homework Solutions

Solution 1. Downsampling a finite sequence

Downsampling by $M = 3$ keeps input indices 0, 3, 6, 9.

$$x_d[n] = \{3, 4, 2, 3\}$$

The output sample rate is $12 \text{ kHz} / 3 = 4 \text{ kHz}$. Discarded indices are 1,2,4,5,7,8,10,11.

Solution 2. Maximum alias-free decimation factor

For alias-free downsampling by M , the preserved input band must satisfy $\omega_{\max} \leq \pi/M$.

$$0.18\pi \leq \pi/M \Rightarrow M \leq 1/0.18 \approx 5.56$$

Therefore the largest integer is $M = 5$. Its ideal aliasing boundary is $\pi/5 = 0.20\pi$. A practical FIR would place its passband edge below 0.20π and include a transition band.

Solution 3. Where does a sinusoid alias?

Downsampling by 2 maps the input digital frequency to $2(0.76\pi) = 1.52\pi$ modulo 2π .

$$1.52\pi - 2\pi = -0.48\pi$$

Because cosine is even in frequency, $\cos(-0.48\pi n) = \cos(0.48\pi n)$. Thus $x_d[n] = \cos(0.48\pi n)$. Once the 0.76π component has folded to 0.48π , the output samples no longer contain information that distinguishes it from a genuine 0.48π input tone; post-filtering cannot undo the aliasing.

Solution 4. Designing a decimator

Block diagram: $x[n] \rightarrow \text{anti-alias lowpass HAA}(z) \rightarrow \downarrow 4 \rightarrow y[n]$.

The ideal boundary is $\pi/4 = 0.25\pi$. Because useful content extends only to 0.18π and interference begins at 0.25π , a reasonable design is passband edge near 0.18π and stopband beginning at or slightly below 0.25π . The transition region is approximately 0.18π to 0.25π .

A symmetric linear-phase FIR preserves relative phase in the passband and has predictable group delay. An IIR or minimum-phase design may be preferable when latency or computational cost is more important than linear phase.

Solution 5. Strided convolution as a rate changer

Stride-1 valid cross-correlation outputs are computed from three-sample windows:

$$y_1 = [0.25(1)+0.5(2)+0.25(0), 0.25(2)+0.5(0)+0.25(-1), \dots]$$

Numerically: $y_{\text{stride1}} = [1.25, 0.25, 0.25, 1.50, 1.25]$. Keeping every second output gives $y_{\text{stride2}} = [1.25, 0.25, 1.25]$.

This is filtering/correlation followed by selecting every second output sample. The kernel $[1/4, 1/2, 1/4]$ is a short smoothing lowpass, so it is qualitatively a plausible anti-aliasing kernel, although its stopband attenuation is modest and may be insufficient for demanding applications.

Solution 6. Upsampling and interpolation gain

For $L = 3$, insert two zeros after each original sample:

$$x_i[n] = \{1, 0, 0, 2, 0, 0, -1, 0, 0, 0, 0, 0\}$$

There are $L = 3$ compressed spectral copies over one 2π interval. The ideal anti-imaging cutoff is $\pi/L = \pi/3$. The interpolation filter should have DC gain $L = 3$.

Reason: for a constant input, zero insertion makes only one out of every three high-rate samples nonzero, reducing the average level by 3. A lowpass with gain 3 restores the original constant amplitude.

Solution 7. Rational sample-rate conversion

20/30 reduces to 2/3, so $L = 2$ and $M = 3$.

Conceptual chain: $x[n] \rightarrow \uparrow 2 \rightarrow H(z) \rightarrow \downarrow 3 \rightarrow y[n]$.

$$\omega_c = \min(\pi/L, \pi/M) = \min(\pi/2, \pi/3) = \pi/3$$

The combined interpolation filter should have DC gain $L = 2$. A polyphase implementation avoids explicitly computing inserted zeros and intermediate samples that would later be discarded, greatly reducing arithmetic and memory traffic.

Solution 8. Transposed convolution and periodic artifacts

With stride 2, ConvTranspose1d can be interpreted as inserting one zero between adjacent input feature samples and then filtering with a learned kernel, subject to padding/output-padding indexing details.

Kernel length 5 is not an integer multiple of stride 2, so neighboring shifted kernels overlap output positions with a periodic pattern; some positions can receive different total contribution counts. This can create amplitude modulation or periodic artifacts.

Mitigations include: (1) upsample/resize followed by a standard stride-1 convolution, (2) choose kernel/stride combinations with more uniform overlap, (3) add explicit lowpass filtering or spectral regularization.

A smooth lowpass-like initialization starts the model with less imaging energy and more even interpolation behavior, but training can still move the weights away from that response, so it is a useful prior rather than a guarantee.

Weekly Quiz — Student Version

Suggested time: 15–20 minutes. Total: 20 points.

1. Multiple choice — Downsampling (2 pts)

Downsampling by $M = 4$ changes the sample rate from 24 kHz to: A) 96 kHz B) 24 kHz C) 6 kHz D) 4 kHz

2. Short answer — Aliasing boundary (2 pts)

For decimation by $M = 3$, what is the ideal maximum input digital frequency that can be preserved without aliasing?

3. Multiple choice — Decimation (2 pts)

Which block order is correct? A) $\downarrow M$ then LPF B) LPF then $\downarrow M$ C) $\uparrow M$ then LPF D) LPF only

4. Short calculation — Frequency mapping (3 pts)

A tone at 0.6π rad/sample is downsampled by 2. What principal output frequency results?

5. Multiple choice — CNN stride (3 pts)

Why can a stride-2 Conv1d alias? A) Every learned kernel is highpass B) The layer samples every second filtered location without guaranteeing enough lowpass suppression C) PyTorch changes the sample amplitudes randomly D) Convolution is nonlinear

6. Short answer — Upsampling (3 pts)

What does upsampling by $L = 4$ insert between adjacent original samples, and what happens to the DTFT spectrum?

7. Short answer — Interpolation gain (3 pts)

Why is the ideal interpolation-filter DC gain L instead of 1?

8. Short calculation — Rational rate (2 pts)

For a rate change $L/M = 3/2$, what ideal combined-filter cutoff should be used on the high-rate intermediate normalized-frequency axis?

Weekly Quiz — Solution Key

1. Answer

C) 6 kHz. The output rate is $24/4 = 6$ kHz.

2. Answer

$\pi/3$ rad/sample. In practice the anti-aliasing passband must end somewhat below this because a realizable filter needs a transition band.

3. Answer

B) LPF then $\downarrow M$.

4. Answer

After $\downarrow 2$, frequency is $2(0.6\pi) = 1.2\pi$. Wrap into $[-\pi, \pi)$: $1.2\pi - 2\pi = -0.8\pi$. For a cosine, this is equivalent to $+0.8\pi$.

5. Answer

B) The layer samples every second filtered location without guaranteeing enough lowpass suppression above the post-stride Nyquist limit.

6. Answer

Insert $L-1 = 3$ zeros between adjacent original samples. The spectrum is compressed by 4 and repeated, producing 4 images/copies over one 2π interval.

7. Answer

Zero insertion reduces the average/DC level of a constant sequence by L . A lowpass with DC gain L restores the original amplitude after interpolation.

8. Answer

$\min(\pi/3, \pi/2) = \pi/3$.

Quiz grading guidance

Response quality	Typical credit	What to look for
Complete	Full	Correct rate/frequency plus the relevant sampling or filtering reason.
Mostly correct	About 70–80%	Right concept with a minor arithmetic, sign, or wording error.
Partial	About 40–60%	Shows the correct operation or formula but not a correct conclusion.
Minimal	0–25%	Unsupported guess or reasoning that reverses anti-alias/anti-imaging order.